

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1924

State Street 230 kV

October 18, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1924. The Interconnection Customer proposes a 573.2 MW natural gas combined cycle facility interconnecting at the State Street 230 kV substation in the Indiana area. The request was received on November 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 681 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,741,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                   | Contingency                                    | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------------------|------------------------------------------------|-------------|--------------|----------|
| State St to Robison Park 230 KV L                    | Swan Valley to Trego 500 KV                    | 123.5       | 2,059        | 41.4     |
| Henry County 500-230 kV XFMR                         | E Methuen to Lonetree Substation 500 KV Line   | 107.2       | 2,099        | 6.2      |
| State Street 230/100kV Transformer Bank              | Robison Park to State Street 230 KV            | 107.2       | 359          | 10.3     |
| State Street 230/100kV XFMR Unit 2                   | State St to Robison Park 230 KV                | 107.2       | 359          | 10.3     |
| State St Substation 230 kV/100 kV Autotransformer #3 | State St to Robison Park 230 KV line           | 107.2       | 359          | 10.3     |
| State St 230 kV/100 kV TX Bank 4                     | State Street to Robison Park 230 KV            | 107.2       | 359          | 10.3     |
| State St to Peavine 100 KV Line 1                    | State St to Robison Park 230 KV                | 104.7       | 276          | 8.8      |
| State St to Peavine 100 KV Line 2                    | State St to Robison Park 230 KV line           | 104.7       | 276          | 8.8      |
| Myrtle Pt to George Neal S Substation 230 KV Line    | Swan Valley to Trego 500 KV                    | 103.1       | 2,123        | 6.1      |
| Acadia to MontourSta 500 KV                          | Coronet to Westwood Generation Llc 500 KV Line | 100.9       | 3,248        | 7.6      |
| Canal Road to Weedpatch 230 KV                       | Trego to Beaur Substation 500 KV               | 100.2       | 1,546        | 7.5      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| State St Substation to Robison Park 230 KV        | Rebuild line with higher-capacity conductor | \$9,065,000   | 11% (\$990,000)       |
| Henry County 500 kV/230 kV Auto                   | Replace transformer with higher-rated unit  | \$39,127,000  | 2% (\$791,000)        |
| State St 230/100 kV Transformer Bank Unit 1       | Replace transformer with higher-rated unit  | \$7,529,000   | 55% (\$4,112,000)     |
| State Street 230/100kV XFMR Unit 2                | Replace transformer with higher-rated unit  | \$7,526,000   | 55% (\$4,111,000)     |
| State St 230-100 kV Transformer                   | Replace transformer with higher-rated unit  | \$10,185,000  | 55% (\$5,564,000)     |
| State St 230/100kV Transformer Bank Unit 4        | Replace transformer with higher-rated unit  | \$9,472,000   | 55% (\$5,174,000)     |
| Peavine to State Street 100 KV Circuit 1          | Reconductor line                            | \$734,000     | 50% (\$371,000)       |
| State St to Peavine 100 KV Line 2                 | Reconductor line                            | \$832,000     | 50% (\$421,000)       |
| Myrtle Pt to George Neal S Substation 230 KV Line | Reconductor line                            | \$6,516,000   | 6% (\$364,000)        |
| Acadia to MontourSta 500 KV                       | Reconductor line                            | \$46,820,000  | 3% (\$1,614,000)      |
| Canal Road to Weedpatch 230 KV                    | Reconductor line                            | \$5,612,000   | 4% (\$229,000)        |
|                                                   | Total allocated to project                  |               | \$23,741,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.